# Lecture 14: Simple and Multiple Linear Regression

IEOR E4709: Statistic Analysis and Time Series
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## 1 Simple Linear Regression

Last lecture, we have introduced simple linear regression and the corresponding estimators.

**Note:** The OLS estimator is a consistent estimator, that is, as $n \to \infty$ the estimators $\hat{\alpha}$ and $\hat{\beta}$ converge, in probability, to the true value of the parameters $\alpha$ and $\beta$, respectively. We will provide a proof in the most general case of multiple linear regression, which also covers the simple linear regression as a special case.

**Remark 1.1.** *The software package R can be used to compute statistics of linear regression models. The R command is lm. For instance, if returns denotes the time series of stock returns and index denotes the time series of index returns lm(returns* $\sim$ *index) regresses the stock returns versus the index returns. Define lm.r = lm(returns* $\sim$ *index). Then, type summary(lm.r) to obtain a summary of the statistics, including estimates of the regression coefficients, standard errors, values of the t statistics, etc...*

### 1.1 Mean Response versus Predicted Response

The mean response at a given value $x_i$ is an estimate of the the mean of the $y$ values in the population when $x = x_i$, that is

$$\hat{y}_i = \hat{\mathbb{E}}\left[y_i | x_i\right] = \hat{\alpha} + \hat{\beta} x_i.$$

The variance of the mean response is given by

$$Var\left(\hat{\alpha} + \hat{\beta} x_i\right) = Var\left(\hat{\alpha}\right) + x_i^2 Var\left(\hat{\beta}\right) + 2 x_i Cov\left(\hat{\alpha}, \hat{\beta}\right),$$

Assume that the $\epsilon_i$'s are i.i.d. Gaussian random variables. Using the expressions for variances and covariances of the least square estimates given above:

$$\begin{aligned} Var(\hat{\beta}) &= S_{xx}^{-1} \sigma^2 \\ Cov(\hat{\alpha}, \hat{\beta}) &= -\bar{x} S_{xx}^{-1} \sigma^2 \\ Var(\hat{\alpha}) &= \frac{\sigma^2}{n} + \bar{x}^2 S_{xx}^{-1} \sigma^2, \end{aligned}$$

along with the identity $\sum_j (x_j - \bar{x})^2 = \sum_j x_j^2 - \frac{1}{n}\left(\sum_j x_j\right)^2$, we can further simplify to

$$Var\left(\hat{\alpha} + \hat{\beta}x_i\right) = \sigma^2\left(\frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}\right). \tag{1}$$

Suppose we observe a new explanatory variable $x_{new}$. We would like to predict the corresponding response $y_{new}$. The variability of this prediction has two sources: the variation in the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$, and the variation due to the fact that $y_{new}$ will not be equal to its expectation, since even if we knew the parameters exactly, the response value will generally not be equal to its expectation because of the error term $\epsilon_{new}$. We have $\hat{y}_{new} = \hat{\alpha} + \hat{\beta}x_{new}$.

Next, compute the variance of the predicted distribution of residuals at the given point $x_{new}$. The distribution of the predicted response is distribution of the predicted residuals at the given point $x_{new}$. Hence, the variance is given by

$$Var\left(y_{new} - (\hat{\alpha} + \hat{\beta}x_{new})\right) = Var(y_{new}) + Var\left(\hat{\alpha} + \hat{\beta}x_{new}\right).$$

The second part of the above expression is given by (1). Since $Var(y_{new}) = \sigma^2$, we get that the variance of the predicted response is given by

$$Var\left(y_{new} - (\hat{\alpha} + \hat{\beta}x_{new})\right) = \sigma^2\left(1 + \frac{1}{n} + \frac{(x_{new} - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}\right). \tag{2}$$

Notice that the variance of the mean response is smaller than the variance of the predicted response, which is intuitive. There is greater uncertainty in predicting one observation (the next observation) than in estimating the mean response given $x_i$. Notice that, as for mean response, the variance of the predicted response also depends on the unknown constant $\sigma^2$. We thus need to replace $\sigma^2$ with its estimate $\hat{\sigma}^2_{rsdl}$, and work with the standard error of the predicted response given by

$$\hat{\sigma}_{rsld}\sqrt{\left(1 + \frac{1}{n} + \frac{(x_{new} - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}\right)},$$

and we use

$$s.e(\hat{y}_i) = \hat{\sigma}_{rsld}\sqrt{\left(\frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}\right)}$$

for the standard error of the mean response.

It follows that the $(1 - \varrho) \times 100\%$ confidence interval for the mean response, i.e., the expected value of $y_i$ given $x_i$, i.e. for $\mathbb{E}[y_i|x_i]$, is given by

$$\hat{y}_i \pm t_{n-2,\frac{\varrho}{2}} \times s.e.(\hat{y}_i).$$

Suppose we are interested in the predicted response, i.e., we want to predict a future value of $y_{new}$ given a new observation $x_{new}$. The prediction interval for $y_{new}$ given a new observation $x_{new}$ is

$$\hat{y}_{new} \pm t_{n-2,\frac{\varrho}{2}} \times s.e.(y_{new} - \hat{y}_{new}).$$

The formula for the prediction interval is similar to that of a confidence interval, except that the variability is higher because there is the additional term "1" in the formula of the standard error in (2). This is expected, because there is more uncertainty around a specific predicted value than around an average.

## 2 Multiple Linear Regression

Consider the following *multiple linear regression* model:

$$y = \boldsymbol{x}^\top \boldsymbol{\beta} + \epsilon,$$

where $\boldsymbol{x}$ is a $p$ dimensional (column) vector of explanatory variables. We will use bold lowercase symbols throughout to denote vectors, and bold uppercase symbols to denote matrices. For instance, if $\boldsymbol{x} = (1, x)^\top$ and $\boldsymbol{\beta} = (\alpha, \beta)^\top$, then we obtain the simple linear regression model. Let us denote by $\boldsymbol{X} := (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n)^\top$ the matrix of explanatory variables, by $\boldsymbol{y} := (y_1, \ldots, y_n)^\top$ the vector of response variables, and by $\boldsymbol{\epsilon} = (\epsilon_1, \ldots, \epsilon_n)^\top$ the vector of noise terms. The vector $\boldsymbol{x}_1 = (1, 1, \ldots, 1)^\top$. We may then write the multiple linear regression with the data set $\boldsymbol{X}$ as

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}.$$

To calculate the least squares estimates, we proceed as in the simple linear regression model. We calculate the first derivative of $S_{rsdl} = \sum_{i=1} \hat{\epsilon}_i^2 = \sum_{i=1}^n \left[ y_i - \hat{\beta}_1 - \hat{\beta}_2 x_{i2} - \hat{\beta}_3 x_{i3} - \cdots - \hat{\beta}_p x_{ip} \right]^2$ with respect to each $\hat{\beta}_j$, $j = 0, 1, \ldots, p$, and set them to zero.

$$\begin{aligned}
\frac{\partial S_{rsdl}}{\partial \hat{\beta}_1} &:= 2 \sum_{i=1}^n \left[ y_i - \hat{\beta}_1 - \hat{\beta}_2 x_{i2} - \hat{\beta}_3 x_{i3} - \ldots \hat{\beta}_p x_{ip} \right] \times (-1) = 0 \\
\frac{\partial S_{rsdl}}{\partial \hat{\beta}_2} &:= 2 \sum_{i=1}^n \left[ y_i - \hat{\beta}_1 - \hat{\beta}_2 x_{i2} - \hat{\beta}_3 x_{i3} - \ldots \hat{\beta}_p x_{ip} \right] \times (-x_{i2}) = 0 \\
\frac{\partial S_{rsdl}}{\partial \hat{\beta}_3} &:= 2 \sum_{i=1}^n \left[ y_i - \hat{\beta}_1 - \hat{\beta}_2 x_{i2} - \hat{\beta}_3 x_{i3} - \ldots \hat{\beta}_p x_{ip} \right] \times (-x_{i3}) = 0 \\
\ldots \quad \ldots \quad \ldots \quad \ldots & \\
\frac{\partial S_{rsdl}}{\partial \hat{\beta}_p} &:= 2 \sum_{i=1}^n \left[ y_i - \hat{\beta}_1 - \hat{\beta}_2 x_{i2} - \hat{\beta}_3 x_{i3} - \ldots \hat{\beta}_p x_{ip} \right] \times (-x_{ip}) = 0.
\end{aligned}$$

Rewriting this in matrix notation, we obtain

$$\boldsymbol{X}^\top \boldsymbol{X} \hat{\boldsymbol{\beta}} = \boldsymbol{X}^\top \boldsymbol{y},$$

or equivalently

$$\hat{\boldsymbol{\beta}} = \left( \boldsymbol{X}^\top \boldsymbol{X} \right)^{-1} \boldsymbol{X}^\top \boldsymbol{y}.$$